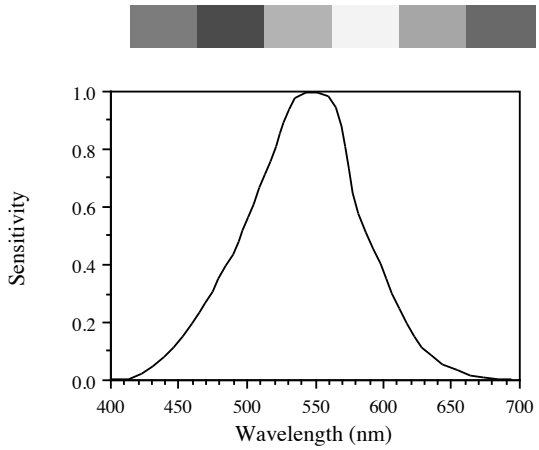
Electromagnetic Fields 2016 – Solutions

1. a) Limitations to atmospheric transmission arise from the three effects of absorption (through electronic transitions at short wavelengths and molecular vibrational transitions at infrared wavelengths), scattering (at short wavelength) and diffraction. The spectral range of sensitivity of the human eye is the so-called visible range, roughly 400 nm < λ < 700 nm. Wavelengths shorter than 400 nm fall into the ultraviolet regime, where there is large electronic absorption. Wavelengths longer than 700 nm fall into the near infrared regime. Although there is little absorption outside specific molecular absorption bands, the photons have too little energy to cause chemical changes that might be easily detectable.

**[4]**

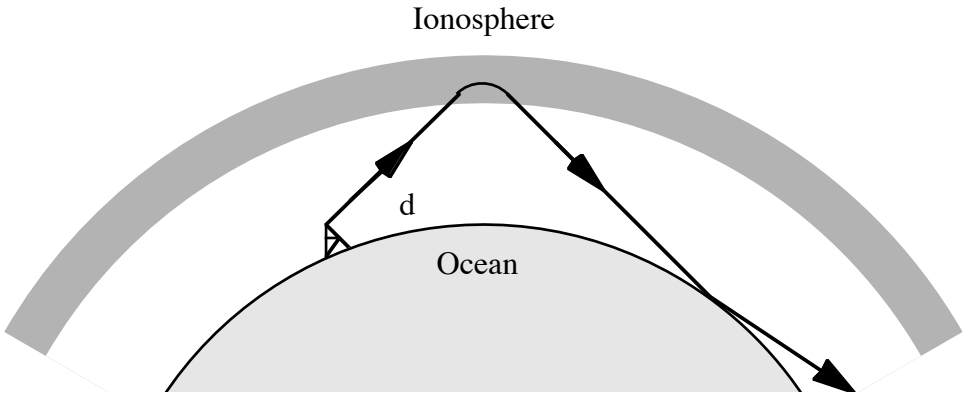



**[4]**

**Answers to this part were mixed. Some students failed to link the discussion of transmission to visible wavelengths relevant to the eye, and instead discussed radio waves. Other students focussed on the physiological details of the eye that had obviously been learnt at A-level, rather than the electromagnetic implications taught in the course.**

b) Marconi's achievement was to demonstrate transatlantic radio communication. Despite the lack of available components, he managed to modulated signals, and then de-modulate them using a simple envelope detector. He achieved over-the-horizon propagation of signals using the ionosphere and the ocean surface as weak reflectors. However, both are sensitive to weather conditions, rendering the communication method relatively unreliable.

**[4]**

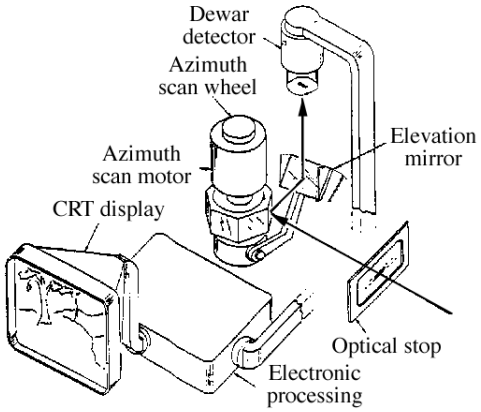



**[4]**

**Most students answered this part well, although some forgot the effect of the ocean.**

c) Night vision systems must detect electromagnetic waves emitted directly by an object, rather than waves from an external source that are scattered by the object. At normal temperatures, the emitted waves are infrared heat waves, which have relatively little energy. They can be detected using band-to-band transitions in a semiconductor with a small bandgap, such as cadmium mercury telluride. However, such transitions are easily generated by thermal energy in the detector so the system must be cryogenically cooled. It is difficult to make a pixellated array of CdHgTe detectors, and consequently the systems often operate by scanning the image over a single detector or a linear array.

**[6]**




**[2]**

**Most students answered this part poorly. Many guessed incorrectly that night vision systems operate using green light, no doubt based of images on green phosphor tubes seen in feature films.**

d) The radar range equation gives the power received by a radar set with transmitter power $P_T$ based on an antenna with efficiency $\eta_T$ and effective area $A_T$ from a target with cross section $\sigma$ at a range r at wavelength $\lambda$. From the Friis transmission equation, the power intercepted by the target is:

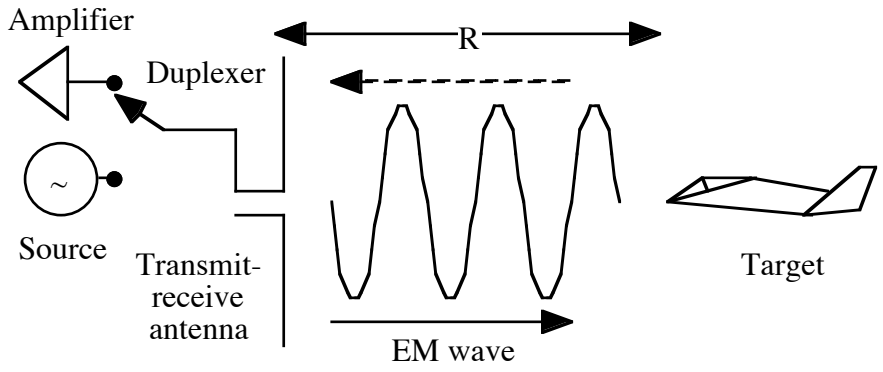
$P_{INT} = P_T(\eta_T A_T \sigma / R^2 \lambda^2)$

Assuming the intercepted power is scattered as a spherical wave, the power received back at the transmit antenna is:

$P_R = P_{INT}\,(\eta_T A_T / 4\pi R^2) = P_T(\eta_T^2 A_T^2 \sigma / 4\pi R^4 \lambda^2)$

For a given detection limit $P_R$, the range then scales as $P_T^{1/4}$, so high transmitter powers are essential for long range.

**[6]**

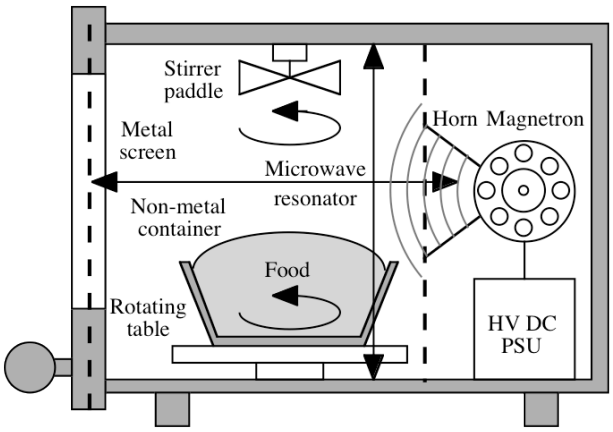



**[2]**

**Generally this question was answered well; however, some students failed to derive the radar range equation correctly, or guess that the received signal falls off as $1/r^4$.**

e) The microwave oven operates by absorption of microwave radiation in one of the molecular absorption bands of the water molecule. A cavity magnetron is used as a source of radiation at 2.45 GHz. Using a horn antenna, energy is coupled into a resonant metal enclosure, which acts as a cooking chamber. Wet food placed in the chamber will then heat up by absorption, but non-uniformly. Uniformity is improved by rotating the food on a table, and by 'stirring' the resonances using a rotating metal paddle.

**[4]**




**[4]**

**Most students answered this question well. The most common error was to provide an extremely poor diagram.**

2. a) The equations linking the voltages and currents $V_n$ and $I_n$ in the $n^{th}$ section to the corresponding values in the $n+1^{th}$ section are:

$V_{n+1} = V_n - j\omega LI_n$

$I_{n+1} = I_n - j\omega CV_{n+1}$

**[2]**

**Most students answered this correctly. The most common error was to present differential equations rather than difference equations.**

Assuming wave solutions $V_n = V_0 \exp(-jnka)$ and $I_n = I_0 \exp(-jnka)$, where k is the propagation constant and a is the length of each section, we get:

$V_0 \exp\{-j(n+1)ka)\} = V_0 \exp(-jnka) - j\omega LI_0 \exp(-jnka)$

$I_0 \exp\{-j(n+1)ka)\} = I_0 \exp(-jnka) - j\omega CV_0 \exp\{-j(n+1)ka)\}$

Cancelling out terms exp(-jnka) on each side, we get:

$V_0 \exp(-jka) = V_0 - j\omega LI_0$

$I_0 \exp(-jka) = I_0 - j\omega CV_0 \exp(-jka)$

Re-arranging, we get:

$\{\exp(-jka) - 1\}V_0 + j\omega LI_0 = 0$

$j\omega C \exp(-jka)\, V_0 + \{\exp(-jka) - 1\}I_0 = 0$

**[4]**

**Most students answered this part correctly, unless their assumed starting point was a pair of differential equations as noted above.**

The result above is in the form $AV_0 + BI_0 = 0$, $CV_0 + DI_0 = 0$.

Consequently, there are only solutions with $V_0 \neq 0$ and $I_0 \neq 0$ if AD - BC = 0, i.e. if:

$\{\exp(-jka) - 1\}\{\exp(-jka) - 1\} + \omega^2 LC \exp(-jka) = 0$

$\exp(-jka) - 2 + \exp(+jka) + \omega^2 LC = 0$

$\omega^2/\omega_0^2 = 2 - 2\cos(ka) = 4\sin^2(ka/2)$

Hence the dispersion relation is $\omega/\omega_0 = 2\sin(ka/2)$

**[4]**

**Most students answered this part correctly, unless their assumed starting point was a pair of differential equations as noted above.**

b) The dispersion relation is clearly sinusoidal. For positive ω, we can plot the dispersion characteristic as the modulus of the result above as shown below.

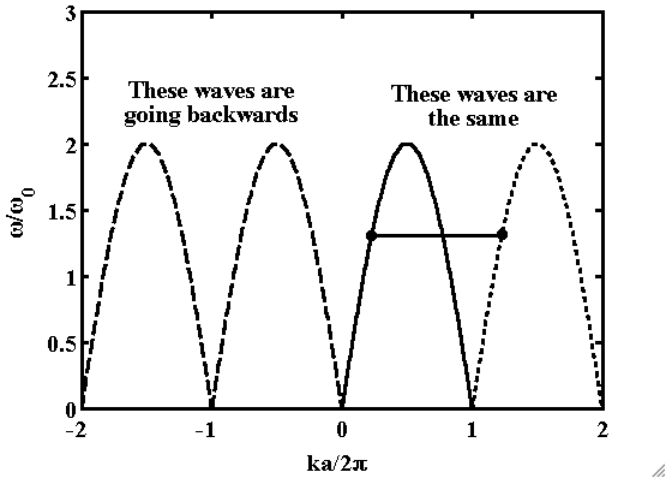
Waves with ka values in the range $0 \le ka \le +2\pi$ represent forward going waves.

Since $V_0 \exp\{-jn(ka + 2\pi)\} = V_0 \exp(-jnka) \exp(-jn2\pi) = V_0 \exp(-jnka)$, the waves in the dotted and full line sections cannot be distinguished from each other.

The waves in the dashed line sections represent backward-going waves.

**[4]**

**Many students failed to notice that some regions of the diagram represent identical waves. A smaller number failed to notice that waves could travel backwards.**




**[6]**

c) The resonant frequency is $\omega_0 = 1/\sqrt{(LC)} = 1/\sqrt{(10 \times 10^{-9} \times 100 \times 10^{-12})} = 10^9$ rad/sec

The cutoff frequency is $\omega_c = 2\omega_0 = 2 \times 10^9$ rad/sec

The value above corresponds to a temporal frequency $f_c = \omega_c/2\pi = 318.3$ MHz

**[2]**

**Many students failed to notice that a frequency must be in Hz.**

If $\omega/\omega_0 = 2 \sin(ka/2)$, ka can be found as $ka = 2 \sin^{-1}(\omega/2\omega_0)$

There is no real solution to $x = \sin^{-1}(y)$ if $y > 1$, the situation when $\omega > 2\omega_0$.

To resolve this, put $y = \sin(x)$ and substitute $x = \pi/2 - jz$.

Then $y = \sin(\pi/2)\cos(jz) + \cos(\pi/2)\sin(jz) = \cos(jz) = \cosh(z)$

Since $z = \cosh^{-1}(y)$ has a real solution for $y > 1$, $x = \pi/2 - j\cosh^{-1}(y)$

**[4]**

Consequently $ka = 2\{\pi/2 - j\cosh^{-1}(\omega/2\omega_0)\}$

Hence, k has a real and an imaginary part and can be written as $k = k' - jk''$.

Wave solutions then have the form $V_0 \exp(-jnka) = V_0 \exp(-jnk'a) \exp(-nk''a)$

Hence the wave decays as it propagates

**[4]**

**Only the best students could do this part. Some students failed to derive expressions for the real and imaginary parts of k; others did not attempt the question at all.**

3a) Ampere's law is:

$\iint_A \underline{\mathbf{D}} \cdot d\underline{a} = \iiint_V \rho \, dv$

Now, Gauss' theorem states that:

$\iint_A \underline{\mathbf{F}} \cdot d\underline{a} = \iiint_V \nabla \cdot \underline{\mathbf{F}} \, dv$

Applying this to Gauss' Law, we get:

$\iint_A \underline{\mathbf{D}} \cdot d\underline{a} = \iiint_V \nabla \cdot \mathbf{D} \, dv = \iiint_V \rho \, dv$

Since the volume is undefined, the last two integrands must be equal. Hence $\nabla \cdot \underline{\mathbf{D}} = \rho$

**[3]**

**Almost all students could do this part.**

Similarly, Faraday's is:

$\int_L \underline{\mathbf{E}} \cdot d\underline{L} = - \iint_A \partial \underline{\mathbf{B}}/\partial t \cdot d\underline{a}$

Now, Stokes' theorem states that:

$\int_L \underline{\mathbf{F}} \cdot d\underline{L} = \iint_A (\nabla \times \underline{\mathbf{F}}) \cdot d\underline{a}$

Applying this to Faraday's Law, we get:

$\int_L \underline{\mathbf{E}} \cdot d\underline{L} = \iint_A (\nabla \times \mathbf{E}) \cdot da = - \iint_A \partial \underline{\mathbf{B}}/\partial t \cdot d\underline{a}$

Since the area is undefined, the last two integrands must be equal.

Hence $\nabla \times \mathbf{E} = - \partial \underline{\mathbf{B}}/\partial t$

**[3]**

**Almost all students could do this part.**

b) i) The time-independent scalar wave equation is:

$\nabla^2 \underline{E} = -\omega^2 \mu_0 \varepsilon_0 \varepsilon_r \underline{E}$

Or:

$\partial^2 \underline{E}/\partial x^2 + \partial^2 \underline{E}/\partial y^2 + \partial^2 \underline{E}/\partial z^2 = -\omega^2 \mu_0 \varepsilon_0 \varepsilon_r \underline{E}$

Assuming the wave is polarized in the x-direction, $\underline{E} = E_x \underline{i}$ and

$\partial^2 E_x/\partial x^2 + \partial^2 E_x/\partial y^2 + \partial^2 E_x/\partial z^2 = -\omega^2 \mu_0 \varepsilon_0 \varepsilon_r E_x$

Assuming the field corresponds to an infinite plane wave propagating in the z-direction:

$\partial E_x/\partial x = \partial E_x/\partial y = 0$

$E_x$ is then a function of z only, so

$d^2 E_x/dz^2 + \omega^2 \mu_0 \varepsilon_0 \varepsilon_r E_x = 0$

**[3]**

**Most students could do this part. The most common error was a failure to justify the neglect of the two derivatives $\partial E_x/\partial x$ and $\partial E_x/\partial y$ properly.**

ii) If $E_x = E_0 \exp(-jk_0 z)$

$dE_x/dz = -jk_0 E_x$

$d^2E_x/dz^2 = -k_0^2 E_x$

Substituting into the wave equation, and assuming $\varepsilon_r = 1$ for free space,

$-k_0^2 E_x + \omega^2\mu_0\varepsilon_0 E_x = 0$

So the propagation constant is:

$k_0 = \omega\sqrt{(\mu_0\varepsilon_0)}$

**[2]**

**Most students could do this part. The only ones who failed to do so appeared to have missed the question completely.**

The corresponding magnetic field can be found from the curl equation:

$\text{curl}(\underline{E}) = -j\omega\mu_0\underline{H}$

which gives

$\underline{H} = (-1/j\omega\mu_0)\ \text{curl}(\underline{E})$

In this case, we have:

$$\text{curl}(\underline{E}) = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ E_x & 0 & 0 \end{vmatrix}$$

Or

$\text{curl}(\underline{E}) = \partial E_x/\partial z\ \underline{j} = -jk_0 E_x\ \underline{j}$

The magnetic field is then:

$\underline{H} = (k_0/\omega\mu_0) E_x\ \underline{j} = (k_0/\omega\mu_0) E_0 \exp(-jk_0 z)\ \underline{j}$

**[3]**

**Most students either did this part correctly, or ignored it.**

If we write this as $\underline{H} = H_y \exp(-jk_0 z)\ \underline{j}$ then

$H_y = (k_0/\omega\mu_0) E_x$

And the characteristic impedance of free space is:

$Z_0 = E_x/H_y = (\omega\mu_0/k_0) = \sqrt{(\mu_0/\varepsilon_0)}$

**[2]**

**Most students either did this part correctly, or wrote down the result from memory.**

c) i) In the upper medium, there must be an incident and a reflected wave. In the lower medium, there must only be a transmitted wave. For TE incidence, both fields must be y-polarised. Hence:

$E_{y1} = E_I \exp[-jk_0n_1(z \sin(\theta_1) - x \cos(\theta_1))] + E_R \exp[-jk_0n_1(z \sin(\theta_1') + x \cos(\theta_1'))]$

$E_{y2} = E_T \exp[-jk_0n_2(z \sin(\theta_2) - x \cos(\theta_2))]$

**[3]**

**Most students could do this part. The ones that failed to do so generally assumed the waves were functions of time, not spatial co-ordinates.**

ii) For the electric field, the boundary condition that must be satisfied is that $E_{y1} = E_{y2}$ on $x = 0$. Hence:

$E_I \exp[-jk_0n_1z \sin(\theta_1)] + E_R \exp[-jk_0n_1z \sin(\theta_1')] = E_T \exp[-jk_0n_2z \sin(\theta_2)]$

**[2]**

**Most students could do this part. The ones that failed to do so generally started off with incorrect expressions for the three waves.**

iii) The only way this equation can be satisfied for all z is if:

$\exp[-jk_0n_1z \sin(\theta_1)] = \exp[-jk_0n_1z \sin(\theta_1')] = \exp[-jk_0n_2z \sin(\theta_2)]$

Hence

$\theta_1 = \theta_1'$ Alhazen's law

$n_1\sin(\theta_1) = n_2 \sin(\theta_2)$ Snell's law

**[3]**

**Most students could do this part, or at least remember Alhazen's and Snell's Laws.**

d) Re-arranging Snell's law, we can obtain:

$\sin(\theta_2) = (n_1/n_2) \sin(\theta_1)$

Hence, $\cos(\theta_2) = \sqrt{\{1 - (n_1/n_2)^2 \sin^2(\theta_1)\}}$

When $(n_1/n_2)^2 \sin^2(\theta_1) > 1$ (i.e., beyond the critical angle) there is no real solution for $\theta_2$

However, it is clear that $\cos(\theta_2)$ must be purely imaginary

In this case, we can write $\cos(\theta_2) = \pm j\alpha$

Taking the negative sign, the field in Medium 2 is

$E_{y2} = E_T \exp[-jk_0 n_2(z \sin(\theta_2) - j\alpha x)]$ or

$E_{y2} = E_T \exp[-jk_0 n_2 z \sin(\theta_2)] \exp(k_0 n_2 \alpha x)]$

This field represents a wave travelling parallel to the boundary, whose amplitude decays exponentially as x becomes negative

**[6]**

**Few students could do this part. The most common error was to focus on the transmission or reflection coefficient, not the wave itself (as asked).**